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Where I Becomes Infinite

Comparing LLM Architectures

GPT, Llama, and Mistral

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Session Overview

What You Will Learn

- The shared decoder-only foundation of GPT, Llama, Mistral
- Key architectural differences: norm type, positional encoding, attention variant
- Model families, sizes, and access models
- How to select the right architecture for a production use case
- Trade-offs: proprietary vs open-weight, quality vs cost vs privacy

Concept at a Glance

- **Duration:** 10 minutes
- **Bloom's Level:** B3 (Apply)
- **Difficulty:** L2 (Intermediate)
- **Domain:** AI Applications
- **Focus:** Architecture comparison and selection

Learning Objectives

By the End of This Concept You Will Be Able To

- **Identify** the shared architectural base and key differences among GPT, Llama, and Mistral
- **Explain** what GQA, RoPE, RMSNorm, and SwiGLU contribute to model efficiency
- **Compare** proprietary API models against open-weight models across cost, privacy, and quality axes
- **Select** an appropriate model family for a given production scenario
- **Interpret** benchmark comparisons to inform model selection decisions

The Common Core

GPT, Llama, and Mistral All Share These Fundamentals

- **Decoder-only Transformer:** Autoregressive generation with causal (left-to-right) self-attention masking
- **Training objective:** Next-token prediction using cross-entropy loss over the full vocabulary
- **Subword tokenization:** BPE-based vocabulary (tiktoken for GPT, SentencePiece BPE or tiktoken for Llama, SentencePiece for Mistral)
- **Pre-LN architecture:** LayerNorm or RMSNorm applied before each sub-layer inside the residual branch
- **LM head with weight tying:** Final linear projection mapping hidden state to vocabulary logits

The Architectural Evolution

Key Changes From GPT-2 to Mistral (2018–2024)

- **GPT-2 (2019):** Baseline decoder-only; learned absolute position embeddings; LayerNorm; GELU; standard MHA
- **GPT-3 (2020):** Scale only (175B parameters); same architecture as GPT-2 but with alternating dense/sparse attention
- **Llama 1 (2023):** RMSNorm, RoPE, SwiGLU, PreNorm; first major open-weight model
- **Llama 2 (2023):** Added Grouped Query Attention (GQA); doubled context to 4K; commercial license
- **Mistral 7B (2023):** GQA + Sliding Window Attention (SWA); matched Llama 2 13B at 7B size
- **Llama 3.1 (2024):** 128K context, 128K vocabulary, tool use; GQA standard

GPT Family: Architecture and Characteristics

Architecture Details

- Decoder-only Transformer; Pre-LN
- **Norm:** LayerNorm (GPT-2/3); details undisclosed for GPT-4
- **Positional encoding:** Learned absolute (GPT-2/3); RoPE suspected in GPT-4
- **Attention:** Standard MHA (GPT-2/3); MoE suspected in GPT-4
- **Tokenizer:** tiktoken (cl100k/o200k)

Access and Deployment

- API-only; weights never released
- Pay-per-token: input cheaper than output
- Instruction-tuned variants (ChatGPT, GPT-4o) via RLHF
- GPT-4o: multimodal (text, image, audio, video)
- Highest quality on general benchmarks as of 2024

GPT Family: Version Timeline

Model	Year	Parameters	Key Advancement
GPT-1	2018	117M	First large autoregressive LM on BooksCorpus
GPT-2	2019	1.5B	Zero-shot transfer; Pre-LN; coherent long text
GPT-3	2020	175B	Few-shot in-context learning; API access only
InstructGPT	2022	1.3B+	RLHF instruction tuning; foundation for ChatGPT
GPT-4	2023	Undisclosed	Multimodal input; bar exam top 10%; MoE suspected
GPT-4o	2024	Undisclosed	Native multimodal (text+image+audio); 2$\times$ faster

Llama Family: Key Architectural Innovations

What Llama Changed vs GPT-2

- **RMSNorm** replaces LayerNorm: 7–15% training speedup
- **RoPE** replaces learned absolute positions: better length generalisation
- **SwiGLU** replaces GELU: improves quality with fewer parameters
- **GQA** (Llama 2+): reduces KV cache by 4–8×

Why These Choices Matter

- Each change was motivated by empirical or theoretical evidence for improvement
- Taken together, these make Llama 3 8B outperform GPT-3 175B on many benchmarks
- Open weights enable the research community to reproduce, verify, and build on results

Llama Family: Version Comparison

Model	Sizes	Context	Vocab	Key Innovation
Llama 1	7B–65B	2K	32K	Open weights, RoPE, SwiGLU, RMSNorm
Llama 2	7B–70B	4K	32K	GQA, RLHF chat, commercial license
Llama 3	8B–70B	8K	128K	Larger vocab, stronger multilingual
Llama 3.1	8B–405B	128K	128K	Tool use, multilingual, 405B flagship
Llama 3.2	1B–11B	128K	128K	Vision models, edge-optimised 1B/3B

Mistral 7B: Maximum Efficiency at Small Scale

Mistral 7B Architectural Features (2023)

- **Grouped Query Attention (GQA):** 8 KV heads, 32 query heads — $4\times$ KV cache reduction
- **Sliding Window Attention (SWA):** Each token attends to the $w = 4{,}096$ most recent tokens; effective context built via stacking across layers
- **Rolling buffer KV cache:** Fixed-size cache using SWA; memory does not grow with sequence length
- **RMSNorm, RoPE, SwiGLU:** Same choices as Llama 2, fully compatible tooling
- **Result:** Mistral 7B outperforms Llama 2 13B on reasoning benchmarks

The Sliding Window Attention is Mistral's distinguishing innovation. By restricting attention span per layer but stacking many layers, the effective receptive field at the last layer reaches $w \times N$ tokens, where N is the number of layers.

Mixtral 8×7B: Sparse Mixture of Experts

Architecture

- 8 FFN "experts" per layer; a router selects top-2 per token
- Total parameters: ≈46.7B; active per token: ≈12.9B
- Same attention architecture as Mistral 7B (GQA + SWA)
- Context: 32K tokens

Why MoE Matters

- Active compute = 7B scale; capacity = 47B scale
- Matches Llama 2 70B performance at 7B inference cost
- Mixtral 8×22B matches GPT-3.5 quality
- MoE is the architecture behind GPT-4 (suspected) and Gemini

Architecture Feature Comparison

Feature	GPT-2/3	Llama 3	Mistral 7B	Mixtral
Architecture	Decoder-only	Decoder-only	Decoder-only	Sparse MoE
Norm	LayerNorm	RMSNorm	RMSNorm	RMSNorm
Position	Learned abs.	RoPE	RoPE	RoPE
Attention	Standard MHA	GQA	GQA + SWA	GQA + SWA
FFN	GELU	SwiGLU	SwiGLU	SwiGLU, 8 experts
Vocab Size	50K / 50K	128K	32K	32K
Weights	Closed	Open	Open	Open

Access Model and Cost Comparison

Model	Access	API Cost	Self-Hosted VRAM
GPT-4o	API only	\$2.50 / \$10 per 1M tok	Not applicable
GPT-4o-mini	API only	\$0.15 / \$0.60 per 1M tok	Not applicable
Llama 3.1 8B	Open weight	Free (self-hosted)	$\approx$16 GB (FP16), 5 GB (4-bit)
Llama 3.1 70B	Open weight	Free (self-hosted)	$\approx$140 GB (FP16), 40 GB (4-bit)
Mistral 7B	Open weight	Free (self-hosted)	$\approx$14 GB (FP16), 4 GB (4-bit)
Mixtral 8$\times$7B	Open weight	Free (self-hosted)	$\approx$90 GB (FP16), 26 GB (4-bit)

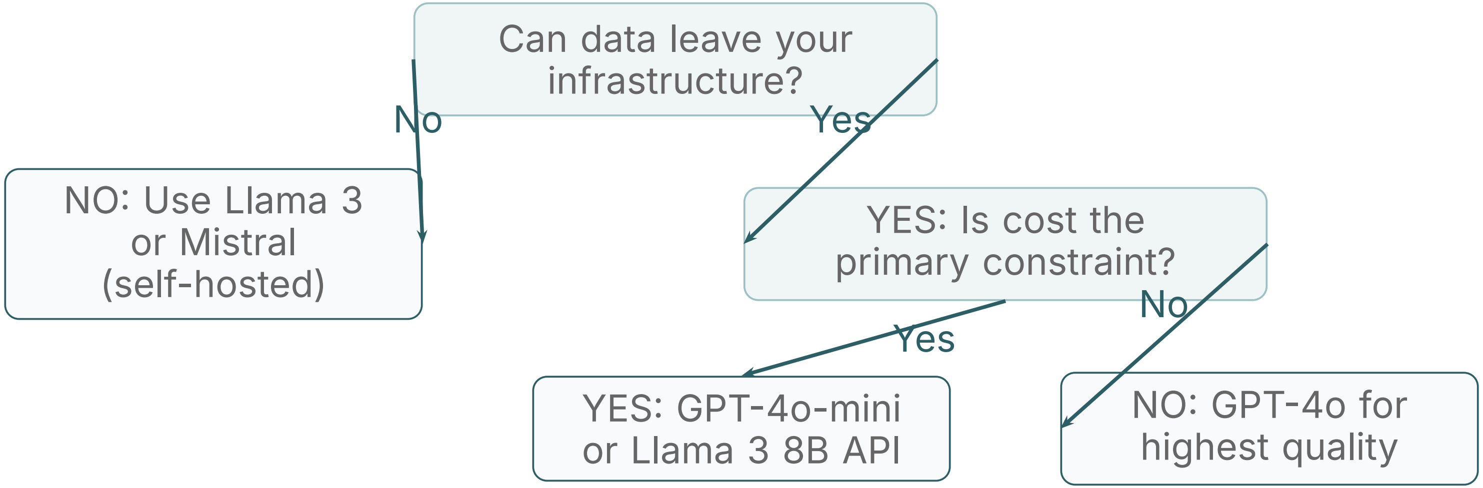
The Model Selection Triangle

Three Competing Objectives

- **Quality:** How well does the model perform on your specific task? (GPT-4o > Llama 3 70B > Mistral 7B on most general benchmarks)
- **Cost:** API costs scale with volume; self-hosted cost depends on GPU infrastructure and engineering
- **Privacy/Control:** Data sent to a proprietary API leaves your infrastructure; self-hosted models keep data entirely on-premises

These three objectives form a triangle: you can optimise for any two but rarely all three simultaneously. Regulated industries (healthcare, finance, legal) often prioritise privacy over quality, leading to Llama or Mistral deployments.

Model Selection Decision Flow



GenAI Application: Enterprise Chatbots

High-Volume Customer Service

- Klarna, Intercom, Freshdesk use GPT-4o via API for quality
- At 100M messages/month, API cost can exceed $1M/month
- Switch to fine-tuned Llama 3 8B when volume justifies infrastructure
- GPT-4o-mini as a cost-quality middle ground

Healthcare and Legal Use Cases

- Patient data and legal documents cannot be sent to OpenAI/Anthropic APIs
- Llama 3 70B or Mistral 7B fine-tuned on domain data
- Deployed on-premises with vLLM or TGI serving
- Epic Systems, Harvey AI use self-hosted approaches

GenAI Application: Code Generation

Specialized Code Models from Each Family

- **GPT-4o / o1-preview (OpenAI):** Highest HumanEval scores; integrated into GitHub Copilot, Cursor, Replit
- **CodeLlama (Meta):** Llama 2 fine-tuned on code; supports infilling (insert code at cursor position); 7B–34B sizes
- **Codestral (Mistral):** 22B parameter code model; optimised for Python, JavaScript, Rust, C++; available via Mistral API and self-hosted
- **Mistral 7B Instruct:** Despite general training, competitive on code tasks within its compute class

For IDE integration with cloud connectivity, GPT-4o provides the best experience. For privacy-sensitive enterprise code review or on-premises deployment, CodeLlama or Codestral are the production-standard choices.

GenAI Application: RAG System Architecture Choices

Matching Model to RAG Role

- **Embedding (retrieval):** Use BERT-family models (not GPT/Llama/Mistral); dedicated embedding endpoints from OpenAI (`text-embedding-3-small`) or open models (E5-Large, GTE)
- **Generation:** GPT-4o for highest-quality synthesis; Llama 3 70B for on-prem; Mistral 7B for fast, cost-effective, lower-stakes RAG
- **Context window trade-off:** GPT-4o supports 128K tokens; Llama 3.1 supports 128K; Mistral 7B supports 32K
- **Prompt caching:** Anthropic and OpenAI cache repeated prefixes; vital for RAG systems with large, static system prompts

Benchmark Comparison (2024)

Model	MMLU	HumanEval	GSM8K	MT-Bench
GPT-4o	88.7%	90.2%	97.0%	9.0
Llama 3.1 70B	86.0%	80.5%	93.0%	8.8
Llama 3.1 8B	73.0%	72.6%	84.5%	8.1
Mistral 7B v0.3	64.1%	38.0%	52.2%	6.8
Mixtral 8×7B	70.6%	40.2%	60.4%	8.3

Common Misconceptions

Myths and Corrections

- **Myth:** “GPT and Llama use fundamentally different architectures.” Both are decoder-only Transformers trained with causal LM; Llama adds engineering improvements, not a new paradigm
- **Myth:** “Open-weight models are free to use.” Weights are free, but GPU infrastructure, electricity, and engineering cost real money; total cost of ownership depends heavily on scale
- **Myth:** “Bigger parameter count always means better quality.” Llama 3 8B outperforms GPT-3 175B on many benchmarks due to better data quality and training recipes
- **Myth:** “Mistral is just a smaller, worse Llama.” Mistral’s Sliding Window Attention and efficient architecture make it uniquely suited for long-context, latency-sensitive applications

Key Takeaways

What You Have Learned

- GPT, Llama, and Mistral all share the decoder-only Transformer with causal LM training; they differ in engineering choices, not fundamental paradigm
- Llama introduced RMSNorm, RoPE, SwiGLU, and GQA over GPT-2 — each improving efficiency and quality
- Mistral added Sliding Window Attention for memory-efficient long-context inference
- GPT models are API-only and provide highest general quality; Llama and Mistral are open-weight for privacy-sensitive, on-premises deployment
- Model selection follows a cost-privacy-quality triangle; the right choice depends on volume, data sensitivity, and quality requirements
- Benchmark performance (MMLU, HumanEval, GSM8K) narrows yearly; Llama 3 70B approaches GPT-4 quality at zero API cost

Thank You

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